

GECON 7201

Applied Econometrics

Assignment 1
Due Thursday October 8, 2026

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Instructions

- Edit this Quarto document for your assignment. All answers must be typed in this document. Answers are to be written in the callout blocks (see below)
- Provide a link to your website in Nexus, so that I can view your assignment.
- Upload both a PDF and Quarto file to the nexus assignment portal.
- The use of AI is strictly prohibited for this assignment.
- Show your work
- Type your mathematical answers using LaTeX
- For questions marked R, include both your code and its output.
- You may use `solve()`, `t()`, `%*%`, `diag()`, `qr()`, `eigen()`, and other base R functions freely unless a question says otherwise.
- Please round numerical answers to two decimal places unless stated otherwise.

```
set.seed(3201)
```

Questions

Q1

Create a personal Quarto website hosted on GitHub pages. For reference, please see the slides for **website** on Nexus. Your website should include the following:

- a. An “Index” or “Home” page. This page should include the following: - Your name - Your photo (optional) - Your credentials - An “About Me” section that provides a brief (one paragraph) introduction to yourself. [10 Marks]

- b. A blog page. This will be the landing site for future assignments [5 Marks]
- c. Create a blog post and upload the finished PDF for this assignment. [3 Marks]
- d. Navigation bar with “Home” and “Blog” buttons. The navigation bar is edited in the “_quarto.yml” file. [2 Marks]

<https://frankshr.github.io/mywebsite/>

Q2

Consider

$$A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 3 & -1 \\ 1 & -1 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 5 & -3 \\ 0 & -3 & 6 \end{bmatrix}.$$

- a. State the dimensions of A and identify a_{23} . [2 Marks]

The dimensions of A are 3×3 . The element in row 2 column 3 is $a_{23} = -1$.

- b. Is A a square matrix? Is it symmetric? Justify your answer using the definition $A = A^\top$, not just by inspection. [2 Marks]

A has 3 rows and 3 columns, so it is a square matrix. When I switch the rows and columns, I get

$$A^\top = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 3 & -1 \\ 1 & -1 & 4 \end{bmatrix}.$$

This is the same as A , so $A = A^\top$. Therefore, A is symmetric.

- c. Is A a diagonal matrix? Explain why or why not. [1 Mark]

A is not a diagonal matrix. A diagonal matrix must have zeros everywhere outside the main diagonal, but $a_{13} = 1$ in A .

- d. In R, Enter A and B into R. Use `isSymmetric()` to confirm your answer to part (b) for both matrices. [4 Marks]

I entered both matrices in R and used `isSymmetric()` to check them.

```
A <- matrix(c(
  2, 0, 1,
  0, 3, -1,
  1, -1, 4
), nrow = 3, byrow = TRUE)
```

```
B <- matrix(c(
  1, 2, 0,
  2, 5, -3,
  0, -3, 6
), nrow = 3, byrow = TRUE)
```

```
isSymmetric(A)
```

```
[1] TRUE
```

```
isSymmetric(B)
```

```
[1] TRUE
```

Both results are TRUE, so both matrices are symmetric. The result for A agrees with my answer in part (b).

Q3

Using the same A and B from Question 1:

- a. Compute $A + B$ and $3A - 2B$ by hand. [4 marks]

$$A + B = \begin{bmatrix} 2+1 & 0+2 & 1+0 \\ 0+2 & 3+5 & -1-3 \\ 1+0 & -1-3 & 4+6 \end{bmatrix} = \begin{bmatrix} 3 & 2 & 1 \\ 2 & 8 & -4 \\ 1 & -4 & 10 \end{bmatrix}.$$

For $3A - 2B$

$$3A - 2B = \begin{bmatrix} 6 & 0 & 3 \\ 0 & 9 & -3 \\ 3 & -3 & 12 \end{bmatrix} - \begin{bmatrix} 2 & 4 & 0 \\ 4 & 10 & -6 \\ 0 & -6 & 12 \end{bmatrix} = \begin{bmatrix} 4 & -4 & 3 \\ -4 & -1 & 3 \\ 3 & 3 & 0 \end{bmatrix}.$$

- b. Compute AB by hand, showing at least one entry's calculation in full using the summation formula $(AB)_{ij} = \sum_{\ell} a_{i\ell}b_{\ell j}$. [2 marks]

I calculate one entry using the summation formula:

$$(AB)_{12} = \sum_{\ell=1}^3 a_{1\ell}b_{\ell 2} = 2(2) + 0(5) + 1(-3) = 1$$

The full result is

$$AB = \begin{bmatrix} 2 & 1 & 6 \\ 6 & 18 & -15 \\ -1 & -15 & 27 \end{bmatrix}.$$

c. Compute BA . Is $AB = BA$ here? Was this guaranteed in general? [2 marks]

$$BA = \begin{bmatrix} 2 & 6 & -1 \\ 1 & 18 & -15 \\ 6 & -15 & 27 \end{bmatrix}.$$

AB and BA are not equal. For example, $(AB)_{12} = 1$, but $(BA)_{12} = 6$. In general, matrix multiplication is not commutative, so $AB = BA$ is not guaranteed.

d. In R, verify parts (a)–(c) in R. Also compute $A * B$ (element-wise) and explain in one sentence why this is a different operation from $A \%*\% B$. [4 marks]

I use R to check my results from parts (a)–(c).

```
A + B
```

```
      [,1] [,2] [,3]
[1,]    3    2    1
[2,]    2    8   -4
[3,]    1   -4   10
```

```
3 * A - 2 * B
```

```
      [,1] [,2] [,3]
[1,]    4   -4    3
[2,]   -4   -1    3
[3,]    3    3    0
```

```
A \%*\% B
```

```
      [,1] [,2] [,3]
```

```
[1,]    2    1    6
[2,]    6   18  -15
[3,]   -1  -15   27
```

```
B %*% A
```

```
      [,1] [,2] [,3]
[1,]    2    6   -1
[2,]    1   18  -15
[3,]    6  -15   27
```

```
A * B
```

```
      [,1] [,2] [,3]
[1,]    2    0    0
[2,]    0   15    3
[3,]    0    3   24
```

`A * B` multiplies entries in the same positions, while `A %*% B` uses row-by-column multiplication and adds the products.

Q4

Let

$$X = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}.$$

- a. Write down $X^\top$ and state its dimensions. [2 marks]

Switching the rows and columns of X gives

$$X^\top = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}.$$

$X^\top$ has 2 rows and 3 columns, so its dimensions are 2×3 .

- b. Compute $X^\top X$ and $XX^\top$ by hand. Explain why these two products have different dimensions even though both involve X and $X^\top$. [4 marks]

Using row-by-column multiplication,

$$X^T X = \begin{bmatrix} 1^2 + 2^2 + 3^2 & 1(4) + 2(5) + 3(6) \\ 4(1) + 5(2) + 6(3) & 4^2 + 5^2 + 6^2 \end{bmatrix} = \begin{bmatrix} 14 & 32 \\ 32 & 77 \end{bmatrix}.$$

Similarly,

$$X X^T = \begin{bmatrix} 1(1) + 4(4) & 1(2) + 4(5) & 1(3) + 4(6) \\ 2(1) + 5(4) & 2(2) + 5(5) & 2(3) + 5(6) \\ 3(1) + 6(4) & 3(2) + 6(5) & 3(3) + 6(6) \end{bmatrix} = \begin{bmatrix} 17 & 22 & 27 \\ 22 & 29 & 36 \\ 27 & 36 & 45 \end{bmatrix}.$$

X is 3×2 and X^T is 2×3 . Therefore, $X^T X$ has dimensions $(2 \times 3)(3 \times 2) = 2 \times 2$, while $X X^T$ has dimensions $(3 \times 2)(2 \times 3) = 3 \times 3$. The result keeps the number of rows of the first matrix and the number of columns of the second matrix.

- c. Let $\mathbf{v} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$. Compute $\mathbf{v}^T \mathbf{v}$ and $\mathbf{v} \mathbf{v}^T$ by hand, and state the dimension of each result.
[4 marks]

For $v^T v$, I multiply the row vector by the column vector:

$$v^T v = \begin{bmatrix} 1 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} = 1^2 + (-1)^2 + 2^2 = 6.$$

The result has dimensions 1×1 .

For vv^T , I multiply the column vector by the row vector:

$$vv^T = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} \begin{bmatrix} 1 & -1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 2 \\ -1 & 1 & -2 \\ 2 & -2 & 4 \end{bmatrix}.$$

The result has dimensions 3×3 .

- d. In R, verify parts (a)–(c). Then confirm numerically, for two conformable matrices of your choosing, that $(AB)^T = B^T A^T$. [4 marks]

I use R to check the transpose, products, and dimensions.

```
X <- matrix(c(
  1, 4,
  2, 5,
  3, 6
), nrow = 3, byrow = TRUE)
```

```
t(X)
```

```
      [,1] [,2] [,3]
[1,]     1     2     3
[2,]     4     5     6
```

```
dim(t(X))
```

```
[1] 2 3
```

```
t(X) %*% X
```

```
      [,1] [,2]
[1,]    14    32
[2,]    32    77
```

```
dim(t(X) %*% X)
```

```
[1] 2 2
```

```
X %*% t(X)
```

```
      [,1] [,2] [,3]
[1,]    17    22    27
[2,]    22    29    36
[3,]    27    36    45
```

```
dim(X %*% t(X))
```

```
[1] 3 3
```

```
v <- matrix(c(1, -1, 2), ncol = 1)
```

```
t(v) %*% v
```

```
      [,1]  
[1,]      6
```

```
dim(t(v) %*% v)
```

```
[1] 1 1
```

```
v %*% t(v)
```

```
      [,1] [,2] [,3]  
[1,]      1    -1     2  
[2,]     -1     1    -2  
[3,]      2    -2     4
```

```
dim(v %*% t(v))
```

```
[1] 3 3
```

For the transpose identity, I use the two matrices from Q2. Both are 3×3 , so their product is defined.

```
t(A %*% B)
```

```
      [,1] [,2] [,3]  
[1,]      2     6    -1  
[2,]      1    18   -15  
[3,]      6   -15    27
```

```
t(B) %*% t(A)
```

```
      [,1] [,2] [,3]  
[1,]      2     6    -1  
[2,]      1    18   -15  
[3,]      6   -15    27
```

```
isTRUE(all.equal(t(A %*% B), t(B) %*% t(A)))
```

```
[1] TRUE
```

The two results are equal, and the check returns TRUE. This confirms $(AB)^T = B^T A^T$ for these matrices.

Q5

Consider

$$A = \begin{bmatrix} 4 & 2 \\ 3 & 3 \end{bmatrix}.$$

- a. In R, compute A^{-1} using `solve()`. [2 marks]

I enter the matrix in R and use `solve()` to find its inverse.

```
A <- matrix(c(
  4, 2,
  3, 3
), nrow = 2, byrow = TRUE)

round(solve(A), 2)
```

```
      [,1] [,2]
[1,]  0.5 -0.33
[2,] -0.5  0.67
```

- b. Construct a 3×3 matrix that is singular (for example, make one column a linear combination of the others) and show what happens when you call `solve()` on it. [2 marks]

```
S <- matrix(c(
  1, 0, 1,
  0, 1, 1,
  1, 1, 2
), nrow = 3, byrow = TRUE)

solve(S)
```

```
Error in `solve.default()`:
! Lapack routine dgesv: system is exactly singular: U[3,3] = 0
```

The columns are linearly dependent, so the matrix is singular. `solve()` returns an error because the inverse does not exist.

Q6

- a. Define, in your own words, what it means for a set of vectors to be linearly independent. [1 mark]

We combine the vectors using coefficients. If the vectors are linearly independent, the combination equals the zero vector only when all coefficients are zero.

- b. Consider

$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{x}_3 = \begin{bmatrix} 2 \\ 5 \\ 3 \end{bmatrix}.$$

Show that $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$ are linearly dependent by finding constants c_1, c_2, c_3 , not all zero, such that $c_1\mathbf{x}_1 + c_2\mathbf{x}_2 + c_3\mathbf{x}_3 = \mathbf{0}$. [3 marks]

I choose $c_1 = 2$, $c_2 = 1$, and $c_3 = -1$. Then

$$2\mathbf{x}_1 + \mathbf{x}_2 - \mathbf{x}_3 = \begin{bmatrix} 2 \\ 4 \\ 2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 2 \\ 5 \\ 3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

These coefficients are not all zero. Therefore, the three vectors are linearly dependent.

- c. Let $M = [\mathbf{x}_1 \ \mathbf{x}_2 \ \mathbf{x}_3]$. What is $\text{rank}(M)$? Explain your reasoning using your answer to part (b). [2 marks]

From part (b), $\mathbf{x}_3 = 2\mathbf{x}_1 + \mathbf{x}_2$. So the third column depends on the first two. $\mathbf{x}_1$ and $\mathbf{x}_2$ are not multiples of each other. They are linearly independent. Therefore, $\text{rank}(M) = 2$.

- d. In R, build M and confirm its rank using `qr(M)$rank`. Then modify one entry of $\mathbf{x}_3$ so that the three vectors become linearly independent, and confirm the new rank. [3 marks]

```
x1 <- c(1, 2, 1)
x2 <- c(0, 1, 1)
x3 <- c(2, 5, 3)

M <- cbind(x1, x2, x3)
qr(M)$rank
```

```
[1] 2
```

The rank is 2. This agrees with part (c).
I change the last entry of $\mathbf{x}_3$ from 3 to 4.

```
x3[3] <- 4  
  
M_new <- cbind(x1, x2, x3)  
qr(M_new)$rank
```

```
[1] 3
```

The new rank is 3. So the three vectors are now linearly independent.

Q7.

Let

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}.$$

- a. Write out the quadratic form $\mathbf{x}^\top A \mathbf{x}$ in terms of x_1 and x_2 . [2 marks]

First, I multiply A by $\mathbf{x}$:

$$A\mathbf{x} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2x_1 + x_2 \\ x_1 + 2x_2 \end{bmatrix}.$$

Then I multiply by $\mathbf{x}^\top$:

$$\mathbf{x}^\top A \mathbf{x} = x_1(2x_1 + x_2) + x_2(x_1 + 2x_2) = 2x_1^2 + 2x_1x_2 + 2x_2^2.$$

- b. Is B invertible? Justify your answer using the relationship between definiteness, rank, and invertibility discussed in class. [3 marks]

For any vector $\mathbf{x}$,

$$\mathbf{x}^\top B \mathbf{x} = x_1^2 + 2x_1x_2 + x_2^2 = (x_1 + x_2)^2 \geq 0.$$

So B is positive semidefinite.

For $\mathbf{x} = (1, -1)^\top$, the quadratic form is zero. This vector is not zero, so B is not positive definite.

The two columns of B are identical and nonzero. Therefore, $\text{rank}(B) = 1 < 2$. B does not have full rank, so it is not invertible.

- c. In R, Confirm parts (a) and (b) by computing `eigen(A)$values` and `eigen(B)$values`. State the rule connecting the sign of a symmetric matrix's eigenvalues to whether it is positive definite, positive semidefinite, or neither. [4 marks]

I enter the matrices from Q7 and check their eigenvalues.

```
A <- matrix(c(
  2, 1,
  1, 2
), nrow = 2, byrow = TRUE)
```

```
B <- matrix(c(
  1, 1,
  1, 1
), nrow = 2, byrow = TRUE)
```

```
eigen(A)$values
```

```
[1] 3 1
```

```
eigen(B)$values
```

```
[1] 2 0
```

Both eigenvalues of A are positive. So A is positive definite. Both eigenvalues of B are nonnegative, but one is zero. So B is positive semidefinite, but not positive definite. The zero eigenvalue also means that B is not invertible.

For a symmetric matrix:

- All eigenvalues are positive: the matrix is positive definite.
- All eigenvalues are nonnegative: the matrix is positive semidefinite.
- At least one eigenvalue is negative: the matrix is neither positive definite nor positive semidefinite.

Q8

Let

$$A = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix}.$$

- a. Show by direct calculation that $A^2 = A$, i.e., that A is idempotent. [2 marks]

I multiply A by itself:

$$A^2 = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix} \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix} = \begin{bmatrix} 0.5(0.5) + 0.5(0.5) & 0.5(0.5) + 0.5(0.5) \\ 0.5(0.5) + 0.5(0.5) & 0.5(0.5) + 0.5(0.5) \end{bmatrix} = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix} = A.$$

Since $A^2 = A$, A is idempotent.

- b. Using only the idempotent property $A^2 = A$ (not this specific matrix), show that $I - A$ is also idempotent. [2 marks]

I expand $(I - A)^2$:

$$(I - A)^2 = I^2 - IA - AI + A^2 = I - 2A + A^2.$$

Since $A^2 = A$,

$$(I - A)^2 = I - 2A + A = I - A.$$

So $I - A$ is also idempotent.

- c. In R, confirm that A is idempotent using matrix multiplication in R. [1 mark]

I use matrix multiplication to check that $A^2 = A$.

```
A <- matrix(c(
  0.5, 0.5,
  0.5, 0.5
), nrow = 2, byrow = TRUE)
```

```
A %*% A
```

```
      [,1] [,2]
[1,]  0.5  0.5
[2,]  0.5  0.5
```

```
isTRUE(all.equal(A %*% A, A))
```

```
[1] TRUE
```

The product equals A . The check returns TRUE, so A is idempotent.

Q9

Let $\mathbf{a} = \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix}$, $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ and

$$S = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 2 \end{bmatrix}.$$

Note that S is symmetric.

- a. Write down $\frac{\partial(\mathbf{a}^\top \mathbf{x})}{\partial \mathbf{x}}$ for a general 3×1 vector $\mathbf{x}$. (No calculation needed, state the rule and apply it to this $\mathbf{a}$.) [2 marks]

The rule is

$$\frac{\partial(\mathbf{a}^\top \mathbf{x})}{\partial \mathbf{x}} = \mathbf{a}.$$

For the given vector,

$$\frac{\partial(\mathbf{a}^\top \mathbf{x})}{\partial \mathbf{x}} = \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix}.$$

- b. Derive $\frac{\partial(\mathbf{x}^\top S \mathbf{x})}{\partial \mathbf{x}}$ using the general rule for $\frac{\partial(\mathbf{x}^\top A \mathbf{x})}{\partial \mathbf{x}}$, and simplify using the fact that S is symmetric. [2 marks]

The general rule is

$$\frac{\partial(\mathbf{x}^\top A \mathbf{x})}{\partial \mathbf{x}} = (A + A^\top) \mathbf{x}.$$

Since S is symmetric, $S^\top = S$. Therefore,

$$\frac{\partial(\mathbf{x}^\top S \mathbf{x})}{\partial \mathbf{x}} = (S + S^\top) \mathbf{x} = 2S \mathbf{x} = \begin{bmatrix} 4x_1 + 2x_2 \\ 2x_1 + 6x_2 + 2x_3 \\ 2x_2 + 4x_3 \end{bmatrix}.$$

- c. Evaluate your answer to part (b) at $\mathbf{x} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$. [1 mark]

I substitute $\mathbf{x} = (1, 0, -1)^\top$ into the result from part (b):

$$2S\mathbf{x} = \begin{bmatrix} 4(1) + 2(0) \\ 2(1) + 6(0) + 2(-1) \\ 2(0) + 4(-1) \end{bmatrix} = \begin{bmatrix} 4 \\ 0 \\ -4 \end{bmatrix}.$$

- d. In R, using the `numDeriv` package (`grad()`), numerically confirm your answers to parts (a) and (c). [2 marks]

I use `grad()` to check both derivatives at $\mathbf{x} = (1, 0, -1)^\top$.

```
a <- c(3, -2, 1)

S <- matrix(c(
  2, 1, 0,
  1, 3, 1,
  0, 1, 2
), nrow = 3, byrow = TRUE)

x0 <- c(1, 0, -1)

f_linear <- function(x) {
  sum(a * x)
}

f_quadratic <- function(x) {
  as.numeric(t(x) %*% S %*% x)
}

round(numDeriv::grad(f_linear, x0), 2)
```

```
[1] 3 -2 1
```

```
round(numDeriv::grad(f_quadratic, x0), 2)
```

```
[1] 4 0 -4
```

The first result matches part (a). The second result matches part (c).

Q10

Let $\mathbf{X} = \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix}$ be a random vector with

$$\mu = E[\mathbf{X}] = \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix}, \quad \Sigma = \text{Var}(\mathbf{X}) = \begin{bmatrix} 4 & 1 & 0 \\ 1 & 2 & -1 \\ 0 & -1 & 3 \end{bmatrix}.$$

Let $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & -1 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 5 \\ -2 \end{bmatrix}$, and define $\mathbf{Y} = A\mathbf{X} + \mathbf{b}$.

- a. State $E[\mathbf{Y}]$ and $\text{Var}(\mathbf{Y})$, citing the properties used. [3 marks]

By the linearity of expectation,

$$E[\mathbf{Y}] = E[A\mathbf{X} + \mathbf{b}] = AE[\mathbf{X}] + \mathbf{b} = A\mu + \mathbf{b}.$$

Adding a constant vector does not change the covariance matrix. For a linear transformation,

$$\text{Var}(\mathbf{Y}) = \text{Var}(A\mathbf{X} + \mathbf{b}) = A \text{Var}(\mathbf{X}) A^\top = A\Sigma A^\top.$$

- b. Compute $E[\mathbf{Y}]$ by hand. [2 marks]

I substitute μ and $\mathbf{b}$ into the formula:

$$E[\mathbf{Y}] = A\mu + \mathbf{b} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix} + \begin{bmatrix} 5 \\ -2 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} + \begin{bmatrix} 5 \\ -2 \end{bmatrix} = \begin{bmatrix} 7 \\ -1 \end{bmatrix}.$$

- c. Compute $\text{Var}(\mathbf{Y}) = A\Sigma A^\top$ by hand, showing your intermediate matrix products. [3 marks]

First, calculate $A\Sigma$:

$$A\Sigma = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} 4 & 1 & 0 \\ 1 & 2 & -1 \\ 0 & -1 & 3 \end{bmatrix} = \begin{bmatrix} 5 & 3 & -1 \\ 1 & 3 & -4 \end{bmatrix}.$$

Then multiply by $A^\top$:

$$\text{Var}(\mathbf{Y}) = A\Sigma A^\top = \begin{bmatrix} 5 & 3 & -1 \\ 1 & 3 & -4 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 5+3 & 3+1 \\ 1+3 & 3+4 \end{bmatrix} = \begin{bmatrix} 8 & 4 \\ 4 & 7 \end{bmatrix}.$$

- d. Let $\mathbf{a} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$. Compute $\text{Var}(\mathbf{a}^\top \mathbf{X})$ using the appropriate property, without recomputing $\text{Var}(\mathbf{X})$ from scratch. [3 marks]

The variance of a linear combination is

$$\text{Var}(\mathbf{a}^\top \mathbf{X}) = \mathbf{a}^\top \Sigma \mathbf{a}.$$

First,

$$\Sigma \mathbf{a} = \begin{bmatrix} 4 & 1 & 0 \\ 1 & 2 & -1 \\ 0 & -1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 \\ -3 \\ 7 \end{bmatrix}.$$

Therefore,

$$\text{Var}(\mathbf{a}^\top \mathbf{X}) = \begin{bmatrix} 1 & -1 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ -3 \\ 7 \end{bmatrix} = 3 + 3 + 14 = 20.$$

- e. In R, using `MASS::mvrnorm()`, draw 100,000 observations from a distribution with mean μ and variance Σ (you may treat $\mathbf{X}$ as multivariate normal for this simulation). Confirm your answers to (b), (c), and (d) using `colMeans()` and `cov()`. [3 marks]

I simulate 100,000 observations and compare the sample results with my calculations.

```

set.seed(3201)

mu <- c(2, 0, -1)

Sigma <- matrix(c(
  4, 1, 0,
  1, 2, -1,
  0, -1, 3
), nrow = 3, byrow = TRUE)

A <- matrix(c(
  1, 1, 0,
  0, 1, -1
), nrow = 2, byrow = TRUE)

b <- c(5, -2)
a <- c(1, -1, 2)

X_sim <- MASS::mvrnorm(
  n = 100000,
  mu = mu,
  Sigma = Sigma
)

Y_sim <- sweep(X_sim %*% t(A), 2, b, FUN = "+")
Z_sim <- X_sim %*% a

round(colMeans(Y_sim), 2)

```

```
[1] 6.99 -1.00
```

```
round(cov(Y_sim), 2)
```

```

      [,1] [,2]
[1,] 8.02 3.99
[2,] 3.99 7.01

```

```
round(cov(Z_sim), 2)
```

```

      [,1]
[1,] 20.09

```

The sample means should be close to 7 and -1. The sample covariance matrix should be close to

$$\begin{bmatrix} 8 & 4 \\ 4 & 7 \end{bmatrix}.$$

The sample variance of $\mathbf{a}^\top \mathbf{X}$ should be close to 20. Small differences are due to random sampling.

Q11

- a. Suppose $\mathbf{X} \sim N_3(\mu, \Sigma)$ using the μ and Σ from Question 10. What is the marginal distribution of X_2 alone? Give its distribution and parameters. [2 marks]

Each component of a multivariate normal vector is normally distributed. For X_2 , the mean is $\mu_2 = 0$ and the variance is $\Sigma_{22} = 2$. Therefore,

$$X_2 \sim N(0, 2),$$

where 2 is the variance. The standard deviation is $\sqrt{2}$.

- c. Using property 4 of the multivariate normal (zero covariance implies independence), determine whether X_1 and X_3 are independent. Justify your answer using an entry of Σ . [1 mark]

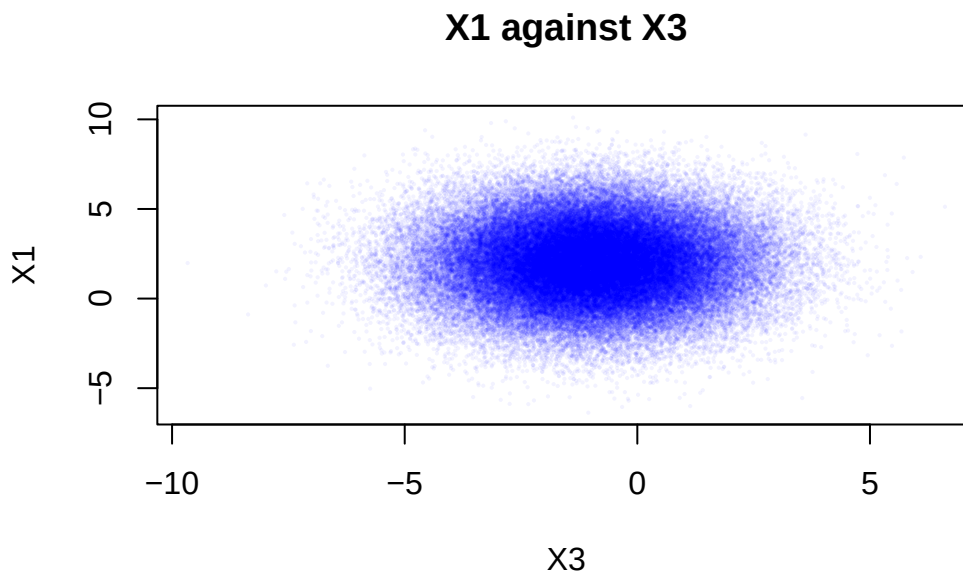
The covariance between X_1 and X_3 is $\Sigma_{13} = 0$. They are jointly normal. For jointly normal variables, zero covariance implies independence. Therefore, X_1 and X_3 are independent.

- d. In R, using your simulated draws from Question 10(e), produce a scatterplot of X_1 against X_3 and compute `cor(X1, X3)`. Comment on whether the plot is consistent with your answer to part (c). [3 marks]

I use the simulated observations from Question 10(e).

```
X1 <- X_sim[, 1]
X3 <- X_sim[, 3]

plot(
  X3, X1,
  xlab = "X3",
  ylab = "X1",
  main = "X1 against X3",
  pch = 16,
  cex = 0.3,
  col = rgb(0, 0, 1, alpha = 0.05)
)
```



```
round(cor(X1, X3), 2)
```

```
[1] 0
```

The plot shows no clear relationship. The correlation is close to zero. This is consistent with part (c). Since the variables are jointly normal, zero population covariance implies independence.